Point of Sale Case Study

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Executive Summary

Super Shopper is facing a lost sales issue across all store locations because of two main inventory management issues: stockout issues and reorder issues. Across all stores, high-demand products are habitually not refilled fast enough and often out of stock. Furthermore, many of these high-demand products are experiencing consecutive weeks with inventory levels of 0. The goal of this analysis is to assess products that are at risk, and estimate the lost sales due to these mismanagements. A secondary concern is low-demand products experiencing little to no movement. These products will also be identified to maximize shelf space distributions. Analysis techniques will be applied to Store 2 so that the process can be repeated for the remaining 9 locations.

Key Variables

Variable	Description
Fully Stocked	The targeted stock level for each product
Reorder Point	The inventory level that when crossed will trigger reorder
Reorder Amount	The quantity of the product that gets ordered when a reorder is triggered
Retail Price	The retail price of the reorder; will be used to calculate lost revenue from stockouts

Identifying Stockouts and Low Inventory Risk

To identify the key problems faced we will highlight any products that meet the following two criteria: any products that had an inventory level of 0 in any given week (Stockout) and any products that dropped below the reorder point but weren't re-stocked (Reorder Issue).

Product	Total Weeks Stocked Out	Total Weeks Not Restocked	Issue
Dish Soap	2	0	Stockout Issue
Dishwasher Soap	3	0	Stockout Issue
Scouring Pads	5	3	Stockout & Reorder Issue
Window Cleaner	3	0	Stockout Issue
Paper Towels	2	0	Stockout Issue
Trash Bags	2	0	Stockout Issue
Toothpaste	2	0	Stockout Issue
Mouthwash	1	0	Stockout Issue
Hand Soap	2	0	Stockout Issue
Shampoo	3	0	Stockout Issue
Hand Lotion	2	1	Stockout & Reorder Issue
Toilet Paper	2	0	Stockout Issue
Cleaning Solvent	0	0	NA
Aspirin	2	0	Stockout Issue
Vitamins	2	0	Stockout Issue
Antacid	0	0	NA
Laxatives	2	1	Stockout & Reorder Issue
Motion Sickness	2	0	Stockout Issue

Pens	1	0	Stockout Issue
Ink Refills	5	2	Stockout & Reorder Issue
Pencils	0	0	NA
Notebooks	4	2	Stockout & Reorder Issue
Paper	0	0	NA
Staplers	3	1	Stockout & Reorder Issue
Paper Clips	1	0	Stockout Issue

Store 2 is experiencing stockout issues for most of its products, and experiencing both stockout issues and reorder issues for an alarming percentage of products as well. Only 4 products in total experienced no stockout issues or reorder issues.

Estimating Lost Sales

Lost sales can be estimated by taking the following steps:

- count the number of weeks a product has at 0 inventory
- calculate the retail price for a single quantity of the product
    - divide the retail price per case by quantity per case
- Multiply the **Fully Stocked** level x **Retail Price** x **Number of Stockout Weeks**

A	B	C	D	E
Product	Total Weeks Stocked Out	Retail Price	Fully Stocked Level	Lost Sales
Shampoo	3	$ 6.60	20	$ 396.00
Aspirin	2	$ 7.20	20	$ 288.00
Staplers	3	$ 9.60	10	$ 288.00
Vitamins	2	$ 6.60	20	$ 264.00
Window Cleaner	3	$ 4.20	20	$ 252.00
Scouring Pads	5	$ 2.40	20	$ 240.00
Notebooks	4	$ 3.60	15	$ 216.00
Paper Towels	2	$ 4.80	20	$ 192.00
Hand Soap	2	$ 4.80	20	$ 192.00
Hand Lotion	2	$ 4.80	20	$ 192.00
Trash Bags	2	$ 4.20	20	$ 168.00
Laxatives	2	$ 8.40	10	$ 168.00
Motion Sickness	2	$ 7.80	10	$ 156.00
Dish Soap	2	$ 3.60	20	$ 144.00
Toothpaste	2	$ 3.60	20	$ 144.00
Dishwasher Soap	3	$ 3.00	15	$ 135.00
Mouthwash	1	$ 4.20	20	$ 84.00
Toilet Paper	2	$ 1.20	20	$ 48.00
Paper Clips	1	$ 4.20	10	$ 42.00
Pens	1	$ 1.80	20	$ 36.00
Ink Refills	5	$ 0.60	10	$ 30.00
Cleaning Solvent	0	$ 5.40	15	$ -
Antacid	0	$ 6.00	20	$ -
Pencils	0	$ 1.20	20	$ -
Paper	0	$ 5.40	15	$ -
Total				$ 3,675.00

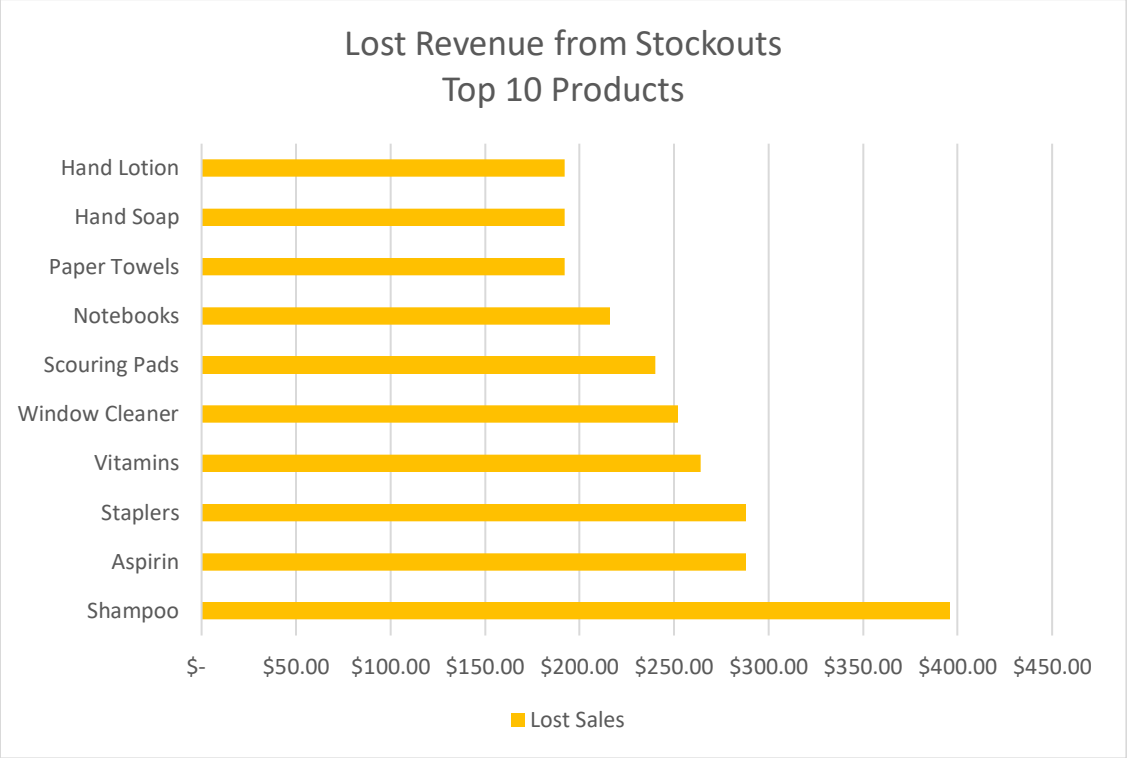
Assuming full demand, Store 2 has a lost sale revenue total of $3,675 over the 8-week sample period. Lost sales are sorted so that the most critical products can be easily identified and prioritized.

Identifying Dead Stock

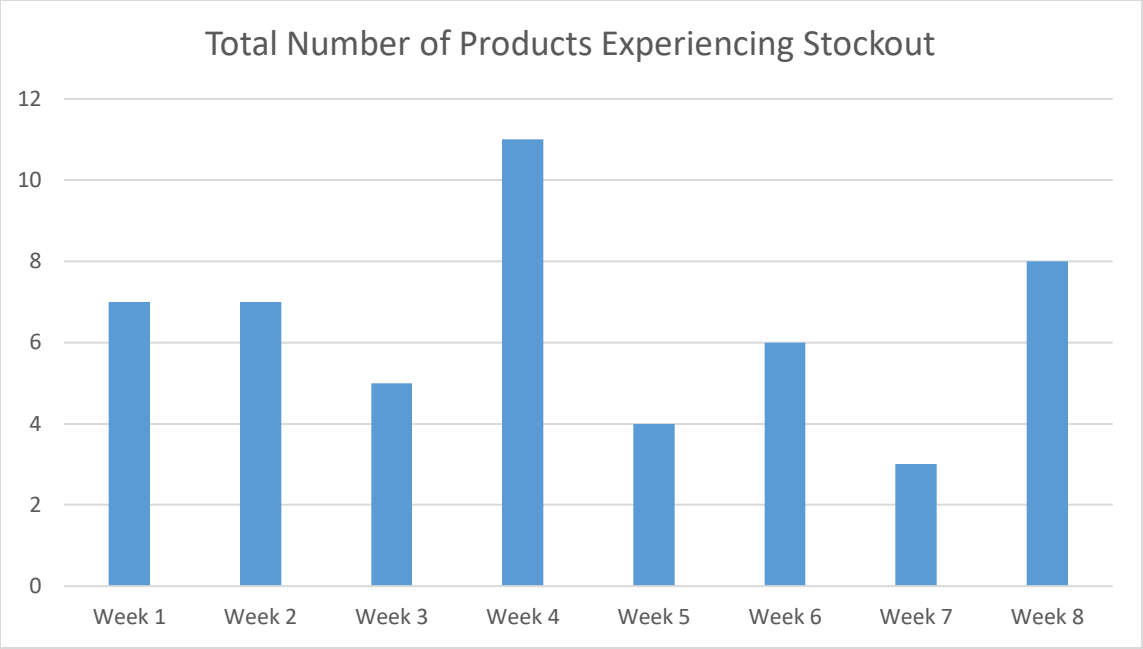
Dead stock is classified as products that are consistently at or near full stock with no decrease over time. These products are not selling and are taking up valuable shelf space. No movement in inventory levels indicates the products have low demand and are overstocked. There are two products (Cleaning Solvent & Pencils) displaying these characteristics at Store 2.

A	B	C	D	E	F	G	H	I	J
		Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Present
	Fully Stocked	Inventory	Inventory	Inventory	Inventory	Inventory	Inventory	Inventory	Inventory
Cleaning Solvent	15	15	15	15	15	15	15	15	15
Pencils	20	20	20	20	20	20	20	20	20

Visualizations & Recommendations



This visualization highlights the top 10 products that result in the highest value of lost revenue for the store. These high-margin items should be prioritized during restock to limit the impact of sales loss. The cost of delayed reordering issues is clearly detailed.



This visualization shows the total number of products, on a week-to-week basis, that Store 2 is stocked out of. Most alarmingly, there is not a single week in the sample where Store 2 avoids any form of stock out. This suggests a significant disconnect between the warehouse reordering procedure and the retail floor. Static reordering procedures should be recalibrated to dynamically address inventory needs on a store-to-store level.